Ground-Effect Model for a Co-Planar Hexarotor at Tilted Attitude

Nomenclature

Symbol	Meaning
R	propeller radius
d	rotor spacing (adjacent rotor axes; equals the arm length for a hexagon)
c	width of the central frame
s	source strength, $s = R^2 v_\infty/4$
v_∞	induced velocity out of ground effect
δv_{ij}	contribution of the image of rotor j to the deficit at rotor i
δv_i	total deficit at rotor i, $\delta v_i = \sum_j \delta v_{ij}$
Δv_f	deficit produced by the fountain flow at the body centre
d_{ij}	horizontal separation between rotor axes i and j
r_{ij}	distance from rotor i to the *image* of rotor j
z_i	height of rotor i above the ground
z_c	height of the body centre above the ground
$h_0,\ l_p$	nominal rotor height at level attitude / pivot offset
J_k	empirical body-lift coefficient
k	matching constant between the two rotor-term constructions
b_i	moment arm about the *body centre*: $l_{y,i}$ (roll) or $-l_{x,i}$ (pitch)

1. Rotor as a potential source

Each rotor is represented as a three-dimensional point source. Mass conservation fixes its strength: the volumetric flow through the disk is $Q = \pi R^2 v_\infty$, and a point source distributes this uniformly over a sphere of radius r, so

$$v(r) = \frac{Q}{4\pi r^2} = \frac{R^2 v_\infty}{4r^2} \implies s = \frac{R^2 v_\infty}{4}, \qquad \phi = -\frac{s}{r}, \qquad \mathbf{v} = \nabla\phi = \frac{s}{r^2}\hat{\mathbf{r}}.$$

The quantity that carries through the derivation is $s/v_\infty = R^2/4$, a purely geometric constant; the induced velocity itself never needs to be known.

2. Ground plane as a mirror image

The no-through-flow condition at $z = 0$ is enforced by placing, for each rotor at height $+z$, an image source of equal strength at $-z$. The image generates an upward flow at the rotor that opposes its own downwash; the ground plane is thereby removed and the free-space solution applies.

3. Isolated rotor

The image lies a distance $2z$ directly below, so the deficit is purely vertical:

$$\frac{\delta v}{v_\infty} = \frac{s}{(2z)^2 v_\infty} = \left(\frac{R}{4z}\right)^2.$$

With $T \propto 1/v_i$ at constant induced power and $v_{IGE} = v_\infty - \delta v$,

$$\frac{T_{IGE}}{T_{OGE}} = \frac{1}{1 - (R/4z)^2}. \tag{2}$$

4. Image geometry and vertical projection

Rotor i sits at $(x_i, y_i, +z_i)$ and the image of rotor j at $(x_j, y_j, -z_j)$, so

$$\Delta\mathbf{r} = \big(x_i - x_j,\ y_i - y_j,\ z_i - (-z_j)\big) = \big(x_i - x_j,\ y_i - y_j,\ z_i + z_j\big).$$

The vertical component is $z_i + z_j$ — a *sum*, because the image lies below the ground plane. Its magnitude follows from Pythagoras,

$$r_{ij} = \sqrt{d_{ij}^2 + (z_i + z_j)^2}, \qquad d_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2,$$

so r_{ij} is the hypotenuse of a right triangle with legs d_{ij} and $z_i + z_j$. The unit vector follows by definition, without differentiation:

$$\hat{\Delta\mathbf{r}} = \frac{\Delta\mathbf{r}}{r_{ij}}, \qquad \cos\theta_{ij} = \hat{\Delta\mathbf{r}} \cdot \hat{\mathbf{e}}_z = \frac{z_i + z_j}{r_{ij}}.$$

The image produces a velocity of magnitude s/r_{ij}^2 along $\hat{\Delta\mathbf{r}}$, but rotor thrust in momentum theory is governed by the flow *normal to the disk*: in-plane components carry no axial momentum flux, and for a symmetric layout they cancel among the images in any case. Only the vertical projection reduces the downwash:

$$\frac{\delta v_{ij}}{v_\infty} = \underbrace{\frac{s}{v_\infty}}_{R^2/4} \cdot \underbrace{\frac{1}{r_{ij}^2}}_{\text{magnitude}} \cdot \underbrace{\frac{z_i + z_j}{r_{ij}}}_{\cos\theta_{ij}} = \frac{R^2}{4} \cdot \frac{z_i + z_j}{r_{ij}^3}.$$

Check. For the self-image ($d_{ii} = 0$, $z_i = z_j = z$) the cosine is unity, $r_{ii} = 2z$, and $(R^2/4)(2z)/(2z)^3 = (R/4z)^2$, recovering Eq. (2).

5. Level attitude: recovering the quadrotor result

With all rotors at a common height, $z_i + z_j = 2z$ and each pair contributes $(R^2/2)\,z/(d_{ij}^2 + 4z^2)^{3/2}$. For a quadrotor, referencing rotor 1, two adjacent rotors lie at d and one diagonal rotor at $\sqrt{2}\,d$:

$$\frac{\delta v_1}{v_\infty} = \left(\frac{R}{4z}\right)^2 + R^2 \frac{z}{(d^2 + 4z^2)^{3/2}} + \frac{R^2}{2} \frac{z}{(2d^2 + 4z^2)^{3/2}},$$

matching Eq. (3) of [Sanchez-Cuevas *et al.*, 2017] coefficient for coefficient.

6. General layout and tilted attitude

Summing over all images gives the total deficit at rotor i:

$$\boxed{\frac{\delta v_i}{v_\infty} = \sum_{j=1}^{n} \frac{\delta v_{ij}}{v_\infty} = \frac{R^2}{4} \sum_{j=1}^{n} \frac{z_i + z_j}{\left[\, d_{ij}^2 + (z_i + z_j)^2 \,\right]^{3/2}}} \tag{1}$$

$$z_i = h_0 \cos\phi + (l_{y,i} + l_p)\sin\phi, \qquad z_c = h_0\cos\phi + l_p \sin\phi.$$

The substitution $z \to (z_i + z_j)/2$ with $4z^2 \to (z_i + z_j)^2$ is not a modelling choice: it is forced by the image geometry of Sec. 4. The expression reduces to Eq. (3) for a level quadrotor and to Eq. (2) when only the self-term is retained. **Every rotor is evaluated at its own height z_i; no averaging is introduced, and the attitude dependence of the model resides entirely in this term.**

7. Fountain effect: lift on the central body

Mechanism. The wake cannot pass through the ground and turns into a radial wall jet. Between opposing rotors these streams collide and reverse upward, impinging on the underside of the central frame and generating an additional upward force. This is why multirotor ground effect exceeds the superposition of isolated rotors.

	Driven by	Acts on
Rotor interference	thrust of rotor j	rotor j
Fountain	total downwash	central frame

Model form. The same construction is applied with the field point at the body centre and the horizontal separation replaced by an effective length r_d:

$$\frac{\Delta v_f}{v_\infty} = \frac{2R^2 J_k\, z_c}{\left(r_d^2 + 4z_c^2\right)^{3/2}}, \qquad r_d = 2d - c.$$

$J_k = 2.2$ for a co-planar hexarotor [Garofano-Soldado *et al.*, 2024, Eq. (8)]. Since a single central frame is involved, this term is evaluated at the single height z_c and is common to all rotors. Neither the factor 2 nor J_k is derived: the potential-flow formulation contains no swirl and cannot represent the shear at the collision plane, so J_k absorbs the recirculation strength and is identified from thrust measurements. Note that z_c varies by only 4.7% over 0–10°, against 17% for the rotor-interference sum.

Two per-rotor gains.

$$\gamma_i^{rot} = \left[1 - k(z_c)\frac{\delta v_i}{v_\infty}\right]^{-1}, \qquad \gamma_i^{full} = \left[1 - k(z_c)\frac{\delta v_i}{v_\infty} - \frac{\Delta v_f}{v_\infty}\right]^{-1}.$$

8. Matching of the two rotor-term constructions

J_k was identified by fitting the complete expression Eq. (9) of [Garofano-Soldado *et al.*, 2024], in which the rotor-interference contribution is evaluated at the mid-points between adjacent propellers,

$$S_{rot}^{(9)}(z) = R^2 \frac{z}{\left(\frac{d^2}{4} + 4z^2\right)^{3/2}} + \frac{R^2}{2}\frac{z}{\left(\frac{21d^2}{4} + 4z^2\right)^{3/2}} + R^2 \frac{z}{\left(\frac{13d^2}{4} + 4z^2\right)^{3/2}} + \frac{R^2}{2}\frac{z}{\left(\frac{7d^2}{4} + 4z^2\right)^{3/2}},$$

whereas Sec. 6 evaluates it at the rotor centres. Because J_k was obtained *together with* $S_{rot}^{(9)}$, substituting a different rotor sum while retaining J_k would displace the total from the measured value. The rotor sum is therefore scaled by

$$k(z_c) = \frac{S_{rot}^{(9)}(z_c)}{\left.\dfrac{\delta v_i}{v_\infty}\right|_{z_1=\cdots=z_n=z_c}}.$$

The denominator is evaluated in a level configuration with all rotors placed at z_c, since Eq. (9) is defined for a single rotor clearance only. This confines k to the geometric difference between the two constructions: were the actual tilted heights used instead, k would absorb the attitude effect and cancel the very leveling torque being modelled. k scales only the rotor terms; the fountain term is used as published.

The two constructions differ by less than 4% throughout, and $k \to 1$ as the vehicle moves away from the ground, since the self-image term then dominates and the choice of evaluation point becomes immaterial:

z [m]	z/R	$\delta v_i/v_\infty$	$S_{rot}^{(9)}$	k
0.15	1.18	0.10255	0.09827	0.9582
0.25	1.97	0.05649	0.05433	0.9617
0.315	2.48	0.04135	0.03989	0.9647
0.50	3.94	0.02015	0.01969	0.9770
1.00	7.87	0.00575	0.00570	0.9922

k is re-evaluated at the body-centre height of each attitude rather than held at its level-attitude value; over 0–10° this changes k by 0.13% and the resulting pivot moment by less than 0.1%. Holding k fixed across attitude — as opposed to across height — remains an assumption, since Eq. (9) is stated for a level vehicle only.

Geometric remark. For the hexagonal layout the rotor-to-mid-point distances take three values, $d/2$, $\sqrt{7}\,d/2$ and $\sqrt{13}\,d/2$, each occurring twice, so that a geometrically consistent form of the rotor terms reads

$$S_{rot} = R^2 z \left[\frac{1}{\left(\frac{d^2}{4} + 4z^2\right)^{3/2}} + \frac{1}{\left(\frac{7d^2}{4} + 4z^2\right)^{3/2}} + \frac{1}{\left(\frac{13d^2}{4} + 4z^2\right)^{3/2}} \right].$$

The values reported here are nevertheless obtained with Eq. (9) as published, since J_k was identified together with that expression and the two must be used as a pair. The difference is 7.5% in S_{rot}, i.e. $\eta = 12.04\%$ against 11.67% at h_0.

9. Mapping to the pivot moment

The interference term is attached to a rotor and enters the moment about the body centre through its own arm. The fountain force acts at the geometric centre and therefore contributes *no* moment about that centre; it enters only through the parallel-axis transfer to the pivot, $M_{pivot} = M_{centre} + F\,l_p$:

$$\boxed{\Delta M_{GE} = \underbrace{\sum_i b_i \left(\gamma_i^{rot} - 1\right) T_i}_{\text{moment about the body centre}} + \underbrace{\sum_i \left(\gamma_i^{full} - 1\right) T_i}_{\Delta f_{GE},\ \text{total vertical force}} \cdot l_p} \tag{2}$$

with $b_i = l_{y,i}$ for roll and $-l_{x,i}$ for pitch — arms measured about the body centre, *not* including l_p. Each gain appears where it belongs: γ_i^{rot} in the centre moment, from which the body lift is structurally absent, and γ_i^{full} in the total vertical force, to which both contributions add. Both sums run over rotors with per-rotor values, so no averaging enters. At level attitude all γ_i^{full} coincide and $\Delta f_{GE} = (\gamma^{full} - 1)f$ reproduces the identified thrust ratio exactly.

Summary of where each height is used.

Quantity	Height
δv_i (both gains)	z_i, per rotor
Δv_f (fountain)	z_c, body centre
k (matching)	z_c, level configuration

Alternative distribution of the body lift. The above assigns the fountain force entirely to the geometric centre. [Sanchez-Cuevas *et al.*, 2017, Eq. (10)] instead applies the full ratio as a per-rotor factor, distributing the body lift in proportion to rotor thrust and giving $\Delta M_{GE} = \sum_i (b_i + l_p)(\gamma_i^{full} - 1)T_i$. The two treatments bracket the physical behaviour: recirculation is stronger beneath the more heavily loaded rotors, but does not track them proportionally. Both are reported.

10. Affine decomposition under constant total thrust

With the total thrust held fixed, the commanded wrench splits as $\mathbf{T} = \mathbf{T}_f + \mathbf{T}_M M_{cmd}$ and the ground-effect moment is affine in M_{cmd}:

$$\Delta M_{GE}(\phi) = a(\phi) + b(\phi)\, M_{cmd}.$$

For the present platform ($R = 0.127$ m, $d = 0.265$ m, $c = 0.22$ m, $h_0 = 0.315$ m, $l_p = 0.140$ m, $f = 0.70W$):

ϕ [deg]	k	$a/(fl_p)$ [%]	b [%]
0	0.96465	11.666	4.155
2	0.96495	11.283	3.995
5	0.96536	10.786	3.793
7	0.96561	10.501	3.680
10	0.96593	10.132	3.538

$$a/(fl_p)\ [\%] = 11.588 - 0.153\,\phi, \qquad b\ [\%] = 4.117 - 0.061\,\phi \qquad (\phi \text{ in deg}).$$

The constant term carries the fountain contribution and is therefore the larger of the two; b reflects rotor-to-rotor interference alone, since the body lift does not scale with the commanded moment. Both are independent of the magnitude of f and M_{cmd} under the stated normalization.

Caveats

Source-strength convention. The source defines $s = R^2 v_{IGE}/4$ while proceeding in terms of $\delta v/v_\infty$. Strictly following the former yields $T_{IGE}/T_{OGE} = 1 + (R/4z)^2$; the two agree to first order and differ by 0.01%p at h_0. Keep v_∞ as the reference consistently.

Scope of the potential-flow layer. The image method captures rotor-to-rotor interference and its attitude dependence analytically. It does not represent wake recirculation (hence the empirical J_k), rotor swirl and its direction, or blade geometry. The boundary between the derived and the fitted layer should be stated explicitly.